

Fine numbers are not log-concave

A counterexample to Conjecture 00000003327

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Abstract

The positive consecutive Fine numbers $F_2 = 1$, $F_3 = 2$, $F_4 = 6$ violate log-concavity: $F_3^2 = 4 < 6 = F_2F_4$. We derive these values from hill-free Dyck paths and supply a self-contained Lean 4 verification of the counting definitions and the strict inequality.

1 The claim and its interpretation

Conjecture 00000003327 asserts log-concavity of the Fine number sequence, in the context of group gradings and a proposed Narayana convolution explanation. We use the standard Fine numbers indexed from $F_0 = 1$: F_n counts hill-free Dyck paths of semilength n . This interpretation is stated explicitly in the introduction of Barucci et al. [1].

A Dyck path is a word in $U = (1, 1)$ and $D = (1, -1)$ starting at height zero, never attaining negative height, and ending at height zero. A *hill* is a consecutive UD starting at height zero, thus forming a peak of height one. A path is hill-free if it contains no such pair.

Log-concavity of a nonnegative sequence requires

$$F_n^2 \geq F_{n-1}F_{n+1}$$

at each interior index. We disprove even the weaker requirement restricted to $n \geq 3$, entirely within the positive tail. The zero F_1 plays no role. Shifting the indexing convention also leaves the consecutive triple 1, 2, 6 and its violation unchanged. The conjecture specifies no alternative normalization, no different sequence of graded dimensions, and no eventual-only qualification. Our conclusion concerns its stated Fine-number log-concavity clause; falsifying that clause is sufficient, independently of the suggested proof mechanism.

2 Exact counterexample

Proposition 1. *The Fine number sequence is not log-concave, even on its positive tail.*

Proof. The complete hill-free lists for semilengths 2, 3, and 4 are:

n	Hill-free Dyck paths	F_n
2	UUDD	1
3	UUUDDD, UUDUDD	2
4	UUUUDDDD, UUUDUDDD, UUDDUDD UUDUUDDD, UUDUDUDD, UUDDUUDD	6

To see that the lists are exhaustive, split a path at each return to height zero. A hill-free irreducible component has the unique form UAD , where A is a nonempty Dyck path. For semilengths 2 and 3 there can only be one component, giving respectively the one and two Dyck interiors of semilength 1 and 2. For semilength 4, a single component gives the five Dyck interiors of semilength 3; two components must both have semilength 2, giving one further path. Therefore $F_2 = 1$, $F_3 = 2$, $F_4 = 6$, and $F_3^2 = 4 < 6 = F_2F_4$. $\square$

3 Formal verification and reproducibility

The accompanying `lean4/` project uses Lean 4.19.0 and its standard library only. In `Main.lean`, Boolean lists encode step words, with `true` for U and `false` for D .

- `words` enumerates every word of a prescribed length. Theorems `mem_words` and `words_nodup` prove completeness and absence of repetitions for every length.
- `DyckFrom` recursively states that the path stays nonnegative and ends at zero. `NoHillsFrom` forbids a UD pair at height zero. The equivalence `hillFreeFrom_iff` verifies that the Boolean scanner accepts exactly their conjunction.
- `paths n` filters words of length $2n$ by that scanner. `mem_paths_iff` identifies precisely the admissible paths, and `paths_nodup` guarantees that length counts distinct paths. Thus `fine n := (paths n).length` implements the definition of F_n .
- `fine_two`, `fine_three`, and `fine_four` calculate the required values by kernel-checked `decide`. `violation` proves the strict inequality, and `not_logConcavePositiveTail` negates the universal claim for $n \geq 3$. Finally, `conjecture_00000003327_false` negates the full log-concavity clause for $n \geq 1$.

The printed axiom audit reports no axioms for the three numerical equalities, the strict inequality, or the final negation. The general enumeration/semantics helpers use only Lean’s standard logical axioms (`propext`, `Quot.sound`, and, for the semantics equivalence, `Classical.choice`). There are no added axioms, admitted proofs, or native-evaluation proof shortcuts.

From the submission directory, run:

```
cd lean4
lake build
lake env lean Main.lean
cd ..
python verify.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

Alternatively, `tectonic report.tex` builds the PDF. The independent Python script uses signed prefix heights to enumerate all binary step words. It also checks the first-return recurrence

$$F_0 = 1, \quad F_n = \sum_{k=2}^n C_{k-1} F_{n-k} \quad (n \geq 1), \quad C_j = \frac{1}{j+1} \binom{2j}{j},$$

against the enumeration for $0 \leq n \leq 7$. Python output is supplementary; the Lean contradiction does not depend on it or on imported numerical data.

References

- [1] E. Barcucci, E. Pergola, R. Pinzani, and S. Rinaldi, *ECO method and hill-free generalized Motzkin paths*, Séminaire Lotharingien de Combinatoire **46** (2001), Article B46b, p. 1.
<https://www.mat.univie.ac.at/~slc/wpapers/s46rinaldi.pdf>.